

Multimodal Machine Learning Framework for Real-Time Synthetic Identity Fraud Detection in Digital Lending Onboarding

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Abstract— Digital lending platforms rely on automated onboarding to approve loans rapidly. However, this introduces exposure to **Synthetic Identity Fraud (SIF)**—a financial crime where perpetrators construct composite identities combining real PII fragments with fabricated contact details. Rule-based KYC systems fail to catch synthetic identities because individual field queries return clean matches. This paper presents a multimodal machine learning framework unifying **KYC entity matching, identity freshness, behavioral biometrics, and device velocity** across 20 signals. Benchmark results across 5 classifiers using 5-Fold Stratified Cross-Validation demonstrate an **ROC-AUC of 0.9061** and **Fraud Precision of 0.9562**.

I. Introduction & Mathematical Threat Formulation

Synthetic Identity Fraud (SIF) represents the fastest-growing category of financial crime in digital lending onboarding. Perpetrators construct composite identities defined formally as $S = P_{\text{real}} \cup P_{\text{fabricated}}$. To counter class imbalance (~16% fraud), models utilize class-weighted loss functions $w_j = N / (2 * N_j)$.

II. 5-Fold Cross-Validation Benchmark Results (20 Signals)

Classifier Architecture	ROC-AUC	PR-AUC	Precision	Recall	F1-Score
Logistic Regression	0.9139	0.8476	0.9602	0.8197	0.8844
Decision Tree	0.9000	0.8272	0.8866	0.8185	0.8506
Random Forest	0.9101	0.8522	0.9616	0.8197	0.8849
Gradient Boosting	0.9115	0.8523	0.9594	0.8179	0.8830
Multi-Layer Perceptron (MLP)	0.9003	0.8382	0.8618	0.7966	0.8277



